Recap from last lecture

The activity coefficient of a component in a mixture describes its non-ideal mixing:

$$\underline{G} = \underline{G}^{IM} + \underline{G}^{ex} = \sum x_i \underline{G}_i + RT \sum x_i \ln x_i + RT \sum x_i \ln \gamma_i$$

and $$\mu_i = \left(\frac{\partial G}{\partial n_i} \right)_{T,P,n_j} = \mu_i^{IM} + \mu_i^{ex} = \underline{G}_i + RT \ln x_i + RT \ln \gamma_i$$

An activity coefficient model relates the activity coefficient of a species in a mixture to the composition of that mixture

Activity coefficient models must obey the Gibbs-Duhem equation:

$$x_1 d \ln \gamma_1 + x_2 d \ln \gamma_2 = 0$$

The simplest such function is the "one-constant Margules equation": $\underline{G}^{ex} = A x_1 x_2$

Example of using an activity coefficient model

Liquid mixtures of oxygen and argon can be described using the one constant Margules equation. At 83.8K, A = 148 Jmol^{-1}.

(a) Sketch the molar Gibbs energy of mixing of oxygen and argon at 83.8K, as a function of composition.

(b) Calculate the chemical potentials of oxygen and argon in a 0.3 mole fraction oxygen mixture at 83.8K.

(c) Hence estimate the vapour pressure of oxygen at equilibrium with the liquid mixture.

Solution

(a) Call oxygen component 1 and argon component 2:

$$\Delta \underline{G}^{mix} = \Delta \underline{G}_{IM}^{mix} + \underline{G}^{ex} = RT \sum x_i \ln x_i + A x_1 x_2$$

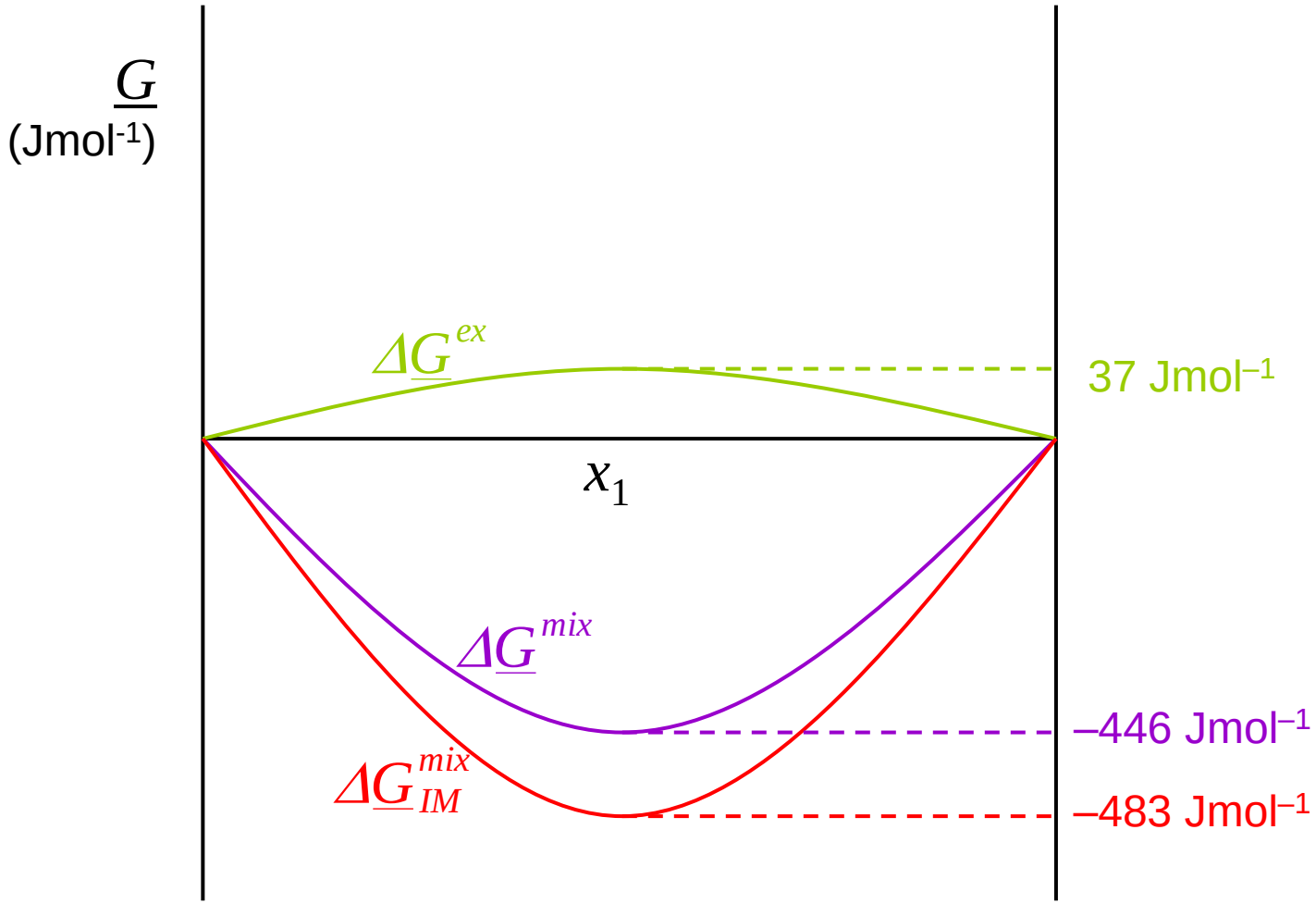
one constant Margules equation

At $x_1 = 0.5$ $\quad \Delta \underline{G}_{IM}^{mix} = -483$ Jmol^{-1}

$\underline{G}^{ex} = 37$ Jmol^{-1}

i.e. the excess term is always small in magnitude relative to the ideal mixing term (for this system)

$\underline{G}^{ex}$ is positive, so we have a positive deviation from ideality – liquid oxygen and argon “do not like” to mix


$\underline{G}$
(Jmol^{-1})
$\Delta\underline{G}^{ex}$
37 Jmol^{-1}
x_1
$\Delta\underline{G}^{mix}$
–446 Jmol^{-1}
$\Delta\underline{G}^{mix}_{IM}$
–483 Jmol^{-1}

(b) Calculate the chemical potentials of oxygen and argon in a 0.3 mole fraction oxygen mixture at 83.8K

$$\underline{G}^{ex} = Ax_1x_2$$ the one constant Margules equation

$$RT\ln\gamma_1 = \mu_1^{ex} = \left(\frac{\partial G^{ex}}{\partial n_1}\right)_{T,P,n_2} = Ax_2^{\,2}$$

(see p4 of Handout 4)

and $$RT\ln\gamma_2 = \mu_2^{ex} = \left(\frac{\partial G^{ex}}{\partial n_2}\right)_{T,P,n_1} = Ax_1^{\,2}$$

$$\mu_i = \mu_i^{IM} + \mu_i^{ex} = \underline{G}_i + RT\ln x_i + RT\ln\gamma_i$$

pure substance, same T,P

ideal mixing term

non-ideal mixing term

$$\mu_1 = \mu_1^{IM} + \mu_1^{ex} = \underline{G}_1 + RT \ln x_1 + A x_2^{\,2}$$

$$= \underline{G}_1 + 8.314 \times 83.8 \ln 0.3 + 148 \times 0.7^2 = \underline{G}_1 - 766 \text{ Jmol}^{-1}$$

$$\mu_2 = \mu_2^{IM} + \mu_2^{ex} = \underline{G}_2 + RT \ln x_2 + A x_1^{\,2}$$

$$= \underline{G}_2 + 8.314 \times 83.8 \ln 0.7 + 148 \times 0.3^2 = \underline{G}_2 - 235 \text{ Jmol}^{-1}$$

Note: Gibbs energy of mixing negative, so mixing is spontaneous

Chemical potential of mixing larger for more dilute species

How would you calculate $\underline{G}_1$ and $\underline{G}_2$?

$$d\underline{G}_1 = -\underline{S}_1 dT + \underline{V}_1 dP$$

(c) Hence estimate the vapour pressure of oxygen at equilibrium with the liquid mixture

For VLE, we require $\mu_1^L = \mu_1^V$

$$\mu_1^L = \underline{G}_1 + RT \ln x_1 + RT \ln \gamma_1$$

pure liquid, same T,P

ideal mixing term

non-ideal mixing term

Assume that the vapour phase behaves as a mixture of ideal gases:

$$\mu_1^V = \underline{G}_1^0 + RT \ln \frac{y_1 P}{P^0}$$

pure IG at T,P^0

Pure liquid oxygen is in equilibrium with pure vapour oxygen at its saturated vapour pressure, $P_1^{SAT}(T)$

Choose $P^0 = P_1^{SAT}$

For a liquid, $\underline{G}_1$ is not a strong fn(P)

so $\underline{G}_1(T, P^{SAT}) \approx \underline{G}_1(T, P)$

$$RT \ln x_1 + RT \ln \gamma_1 = RT \ln \frac{y_1 P}{P_1^{SAT}}$$

$$\boxed{x_1 \gamma_1 P_1^{SAT} = y_1 P}$$ (can you now see why fugacity is convenient?)

$$x_1 \gamma_1 P_1^{SAT} = y_1 P$$

$$x_1 = 0.3$$

$$\ln \gamma_1 = \frac{A}{RT} x_2^{\,2}$$

$$\gamma_1 = \exp\left(\frac{148}{8.314 \times 83.8} 0.7^2 \right) = 1.11$$

We are trying to calculate $y_1 P$

How do we get P_1^{SAT} ?

The boiling point of liquid oxygen is 90.188K, which means its saturated pressure at this temperature is 1 atmosphere

To get the saturated pressure at a different temperature, apply the Clausius-Clapeyron equation

$$\ln \frac{P^{SAT}(T_2)}{P^{SAT}(T_1)} = -\frac{\Delta \underline{H}^{VAP}}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

Clausius-Clapeyron equation

(assuming $\Delta \underline{H}^{VAP}$ not a fn(T))

$$= -\frac{6819}{8.314}\left(\frac{1}{83.8} - \frac{1}{90.188}\right)$$

$$P^{SAT}(83.8\text{K}) = 0.507 \text{ bar}$$

$$x_1 \gamma_1 P_1^{SAT} = y_1 P$$

$$y_1 P = 0.3 \times 1.11 \times 0.507 = 0.169 \text{ bar}$$

We have calculated a (simple) VLE using an activity coefficient model

Using Activity Coefficient Models for Multicomponent Mixtures

Combine binary parameters for each pair of components

e.g. consider simplest case – one constant Margules equation

Binary mixture:

$$\underline{G}^{ex} = Ax_1x_2$$

$$RT\ln\gamma_1 = Ax_2^{\,2}$$

$$RT\ln\gamma_2 = Ax_1^{\,2}$$

from $RT\ln\gamma_i = \left(\frac{dG^{ex}}{dn_i}\right)_{T,P,n_j}$

automatically obeys G-D

For a multicomponent mixture:

$$\underline{G}^{ex} = \sum_{i,j} A_{ij}x_ix_j$$

A_{ij} ← binary parameter

Similar methods for all activity coefficient models

Experimental determination of activity coefficients

Activity coefficient models are fits of an arbitrary (but thermodynamically consistent) function to experimental data

The experimental data we require are activity coefficients as a function of temperature and composition

In principle we need only as many data points as there are free parameters in our model

It is possible to calculate multi-component mixture parameters from binary pair parameters

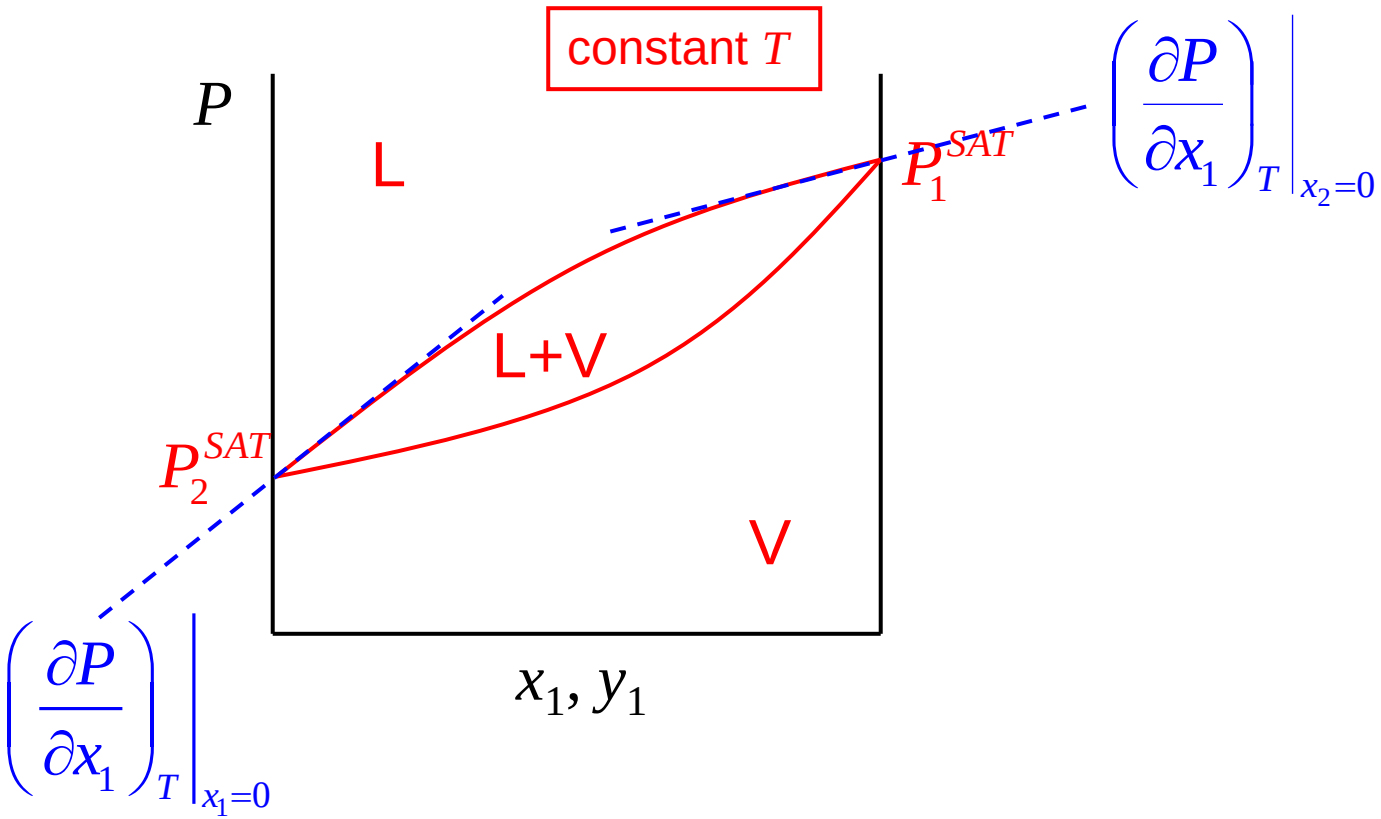
We need to make measurements of activity coefficients for binary mixtures

(i) Infinite dilution ebulliometry

Boil pure component 2 and measure $P = P_2^{SAT}$

Add a small amount of 1. Reboil at same T. Measure ΔP and Δx

Repeat for pure 1, thus estimating $\left(\frac{\partial P}{\partial x_1}\right)_T\Big|_{x_2=0} \approx \left(\frac{\Delta P}{\Delta x_1}\right)_T\Big|_{x_2=0}$



$$dP = \sum \left(\gamma_i P_i^{SAT} dx_i + x_i P_i^{SAT} d\gamma_i \right)$$

$$= \sum \gamma_i P_i^{SAT} \left(dx_i + x_i \frac{d\gamma_i}{\gamma_i} \right)$$

T constant, so only composition varies

$\gamma_i = \text{fn}(x_i)$

binary: $$dP = \gamma_1 P_1^{SAT} \left(dx_1 + x_1 \frac{d\gamma_1}{\gamma_1} \right) + \gamma_2 P_2^{SAT} \left(x_2 \frac{d\gamma_2}{\gamma_2} - dx_1 \right)$$

since $dx_2 = -dx_1$

Apply G-D:

$$dP = \left(\gamma_1 P_1^{SAT} - \gamma_2 P_2^{SAT} \right) \left(dx_1 - x_2 \frac{d\gamma_2}{\gamma_2} \right)$$

at $x_1 = 0, \gamma_2 = 1$ (pure 2) and $\left(\frac{\partial \gamma_2}{\partial x_1}\right)_T = 0$ (from G - D; think about this....)

$$\Rightarrow \left(\frac{\partial P}{\partial x_1}\right)_T \Bigg|_{x_1=0} = \gamma_1^{\infty} P_1^{SAT} - P_2^{SAT}$$

hence $$\gamma_1^{\infty} = \frac{P_2^{SAT} + \left(\frac{\partial P}{\partial x_1}\right)_T \Big|_{x_1=0}}{P_1^{SAT}}$$ and similarly for γ_2^{∞}

We have not had to measure compositions = easy

We can use these two values to obtain the constants for van Laar or a two parameter Margules

(ii) Direct measurements of γ from x-y-P data

(a dynamic still - as opposed to a static still)

P-T-x_i-y_i

boil and separate L + V

measure compositions

P-T-x_i

easier 'cos y_i not measured

$$\gamma_i = \frac{y_i P}{x_i P_i^{SAT}}$$

See p3 this handout (chemical equilibrium assuming IG vapour)

The best approach is to fit parameters from a thermodynamically consistent model (e.g. Margules) on a least squares basis. This can gives very good results. However lots of quite difficult measurements are required because both liquid and vapour compositions are needed.

This is the best approach, because you get lots of data points over a range of T and x_i

However measurements are painful and we need measurable vapour pressures

(iii) Indirect measurements of γ from x-P data (a static still)

Measurements easier and more accurate than dynamic still

$$P = x_1\gamma_1 P_1^{SAT} + x_2\gamma_2 P_2^{SAT}$$ (again assuming vapour phases IG)

activity coefficient model

Fit a little more complex because γ_i appears twice in expression for pressure

Historically the most important method

(iv) Depression of freezing point

Allows activity coefficients to be measured in mixtures where the vapour pressures are very low. See below.

(v) Azeotropes

Allows two activity coefficients (one for each component) to be measured without any need to measure the compositions. See later.

(vi) Osmotic pressure

Allows activity coefficients to be measured in mixtures where the vapour pressures are very low. See later.